Sard's Theorem for Min-Functions according to Yomdin

Albert Fathi

Abstract

Consider a C^k function $F : M^m \times N^n \longrightarrow \mathbb{R}$ defined on the product of the manifold M^m of dimension m with the compact manifold N^n of dimension n.

The function $\inf_{N^n} F : M^m \longrightarrow \mathbb{R}$ defined by is locally Lipschitz. Yomdin proved (in 1984) that if $k \geq k(m, n)$, where the integer $k(m, n)$ depends only on m and n, then for almost every $r \in \mathbb{R}$, the preimage $\{x \mid \inf_{N^n} F(x) = r\}$ is a codimension one submanifold of M^m.

Yomdin's proof relies on his approach to almost critical values, yielding quantitative versions of Sard's theorem.

This bound has subsequently been improved by several authors, always using Yomdin's methods.

We shall give a direct proof based on the classical Morse–Sard theorem.